Tamanjyot Singh

The University of Texas at Dallas, Richardson, TX, USA

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Bio

Aerospace engineer pursuing a **doctorate in physics** with specialization in astrophysics. Interested in understanding **black hole spin** dynamics and their implications. Currently working as a research assistant at the University of Texas at Dallas under the supervision of Dr. Michael Kesden. My thesis work focuses on (i) inclination distributions in **tidal disruption events** by spinning supermassive black holes and their observational implications, (ii) **precessional and nutational** effects on the gravitational wave signals originating from inspiralling stellar mass binary black hole as currently observed by LIGO, and (iii) **degeneracies between lensed and precessing** gravitational wave signals for current LIGO sensitivities.

Education

Ph.D. — **The University of Texas at Dallas** — Richardson, TX, USA
Physics — Jan 2021 - May 2026 *(expected)*
Thesis: *The Black Hole Ballet*: Spin and precession effects on gravitational waves and tidal disruption Events
Advisor: Dr. Michael Kesden

M.S. — **The University of Texas at Dallas** — Richardson, TX, USA
Physics — Jan 2021 - Dec 2023
GPA: 3.93/4.0

B.Tech. — **Punjab Engineering College (Deemed to be University)** — Chandigarh, India
Aerospace Engineering with minor specialization in Mathematics — Aug 2016 - Jun 2020
GPA: 9.2/10.0, Rank 2
Major project: Computational study of flame propagation in a micro-scale channel and applications of micro-combustors in thermo-electric power generation.

Experience

Research assistant — Richardson, TX, USA
The University of Texas at Dallas — May 2022 - Present

- **Lensing and Precession**: Distinguishing between a lensed and precessing gravitational wave signal, exploring degeneracies and establishing differences.
- **Precession in gravitational waves**: Investigating the effects of spin dynamics in a binary black hole system on the gravitational wave emission.
- **Tidal disruption events**: Solving for inclination biases induced by spin breaking the spherical symmetry in stellar disruptions by supermassive black holes.

Responsibilities: Meeting weekly, developing a code, working in a team environment, mentoring students, and preparing manuscripts for publication.

Teaching assistant — Richardson, TX, USA
The University of Texas at Dallas — Jan 2021 - May 2022

- **Physics Laboratory I: Mechanics**: Introductory physics laboratory course for all STEM undergrad tracks.

Responsibilities : Hosting weekly sessions for students to guide them through the mechanics lab experiments. Grading their lab report submissions.

Research Intern — Mumbai, India
Combustion research Laboratory, IIT Bombay — Jan 2019 - Apr 2019

- **Combustion characteristics and effect of hydrogen content on syngas-air mixtures**: Formulated an analytical correlation for laminar burning velocity of multi-component gaseous mixtures simultaneously at elevated pressures and temperatures.

Responsibilities: Preparing weekly progress updates for group meetings. Writing a report and presenting research at Punjab Engineering College, Chandigarh at the end of program.

IAF (Indian Air Force) Intern — Chandigarh, India
3 BRD, Air Force Station — June 2018

Publications

Published:

Distribution of orbital inclinations for tidal disruption events by Kerr black holes — February 2024
Tamanjyot Singh, Michael Kesden
10.1103/PhysRevD.109.043016

Talks

Contributed	**Fall meeting 2024, Texas section APS**, Southern Methodist University, Dallas, TX *Title*: "Signatures of generic precession in gravitational wave signals from merging binary black holes"	October 2024
Invited Visit	Center for Computational Astrophysics, Flatiron Institute, New York City, NY *Title*: "Precession in GWs"	May 2024
Contributed	**APS April meeting 2024**, Sacramento, CA *Title*: "Gravitational wave signals from precessing binary black hole inspirals"	April 2024
Contributed	**TEXAS PAARE Annual Symposium 2024**, University of North Texas, Denton, TX *Title*: "Tidal disruption events: Inclination distributions for spinning supermassive black holes"	March 2024
Contributed	**10th Gulf Coast Gravity meeting 2024**, University of Texas, Austin, TX *Title*: "Gravitational wave signals from precessing binary black hole inspirals"	March 2024
Contributed	**International Conference on Gravitation and Cosmology 2023**, Indian Institute of Technology, Guwahati, Assam, India *Title*: "Inclination distributions in tidal disruption events by spinning supermassive black holes"	December 2023
Contributed	**Fall meeting 2023, Texas section APS**, Angelo State University, San Angelo, TX *Title*: "Identifying Precession in Gravitational Wave signals from Binary Black Hole inspirals"	October 2023
Contributed	**Spring meeting 2023, Texas section APS**, Texas A&M University, Commerce, TX *Title*: "Tidal disruption events by spinning supermassive black holes"	March 2023

Achievements

APS April meeting DGRAV Travel Award — April 2024

Margie Renfrow travel award by the Department of Physics, The University of Texas at Dallas, TX — March 2024

Teaching Assistant award 2022 at The University of Texas at Dallas, TX — December 2022

PECOSA Sydney Prize for year 2017-18 (Merit Based - Awarded to 3 out of 700+ students) — September 2018

Secured **1st position** in the Mathematics Quiz sponsored by Punjab State Council for Science and Technology, Chandigarh (Gold Medal - District Level Competition) — September 2016

99.3%ile in JEE (Mains) 2016 (among 1,500,000+ candidates all over India) — April 2016

International Rank 76 in International Olympiad of Science 2013 (Gold Medal + *Special Achievement Certificate*) — November 2013

Workshops and Course Certificates

Attended **Entering Mentoring** at the University of Texas at Dallas — July 2024

Course Certificate for **Astro 101 : Black Holes** by University of Alberta on Coursera. — June 2020

Hosted a workshop at undergrad level for **Gravitational lensing** at Punjab Engineering College, Chandigarh — October 2019

Attended **ASTROWIN**, Winter School on Astronomy 2019 at the Birla Museum, Hyderabad — February 2019

Course Certificate for **Particle Physics: an Introduction** by University of Geneva on Coursera. — July 2018

Course Certificate for **Understanding Einstein: The Special Theory of Relativity** by Stanford University on Coursera. — July 2017

Mentoring

I have had a chance to be secondary mentor to many undergraduate and near-peer mentees over the years. Following is a list with names and roles of mentees:

Tien Nguyen	PAARE fellow 2023-24 *(Post-baccalaureate student)*
Maya Basu	Summer REU student 2024 *(Undergrad Student)*
Tinatin Tsiskaridze	Summer exchange student 2024 *(Grad Student)* (from Tblisi State University, Georgia)
Niel Shah	Summer REU student 2023 *(Undergrad Student)*
Deven Linthicum	Summer REU student 2023 *(Undergrad Student)*
Allison Powell	Summer REU student 2022 *(Undergrad Student)*

Extra Curricular Responsibilities

Journal Club host: Hosting weekly meetings with students across DFW in a joint astrophysics journal club with University of North Texas, Denton. Graduate students encouraged to present a recent paper in their field of research (of their choice) at least once per semester. — August 2024 - Ongoing

Graduate Students in Physics (GSP) Vice President: Representing graduate student interests in the department. Responsible for assisting President in maintenance and workings of the organization. *Standout accomplishment*: Department-wide Overleaf license. — May 2023 - July 2024

Graduate Students in Physics (GSP) Astronomy committee head: Grad student liaison for all things telescope, hosting star-watch and lunar eclipse parties. Responsible for maintenance of the existing equipment and inventory along with buying new equipment. — March 2022 - February 2024